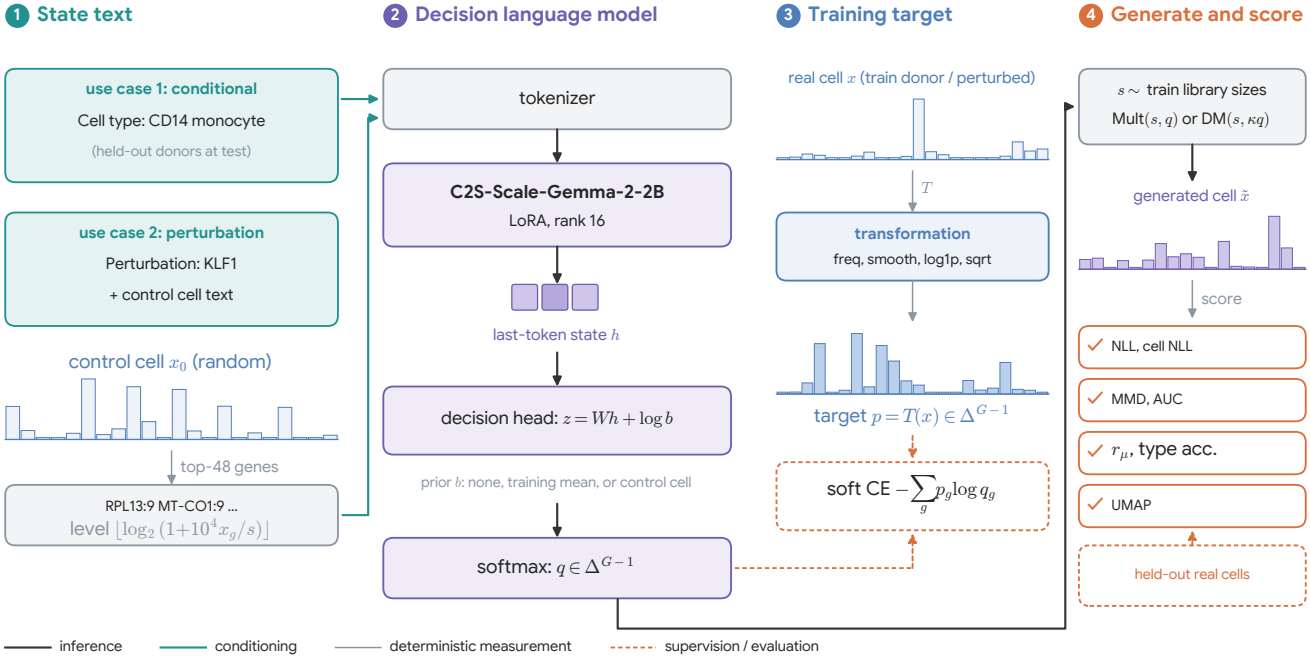
1 State text
2 Decision language model
3 Training target
4 Generate and score
use case 1: conditional
Cell type: CD14 monocyte
(held-out donors at test)
use case 2: perturbation
Perturbation: KLF1
+ control cell text
control cell x_0 (random)
top-48 genes
RPL13:9 MT-CO1:9 ...
level $\lfloor \log_2(1+10^4 x_g/s) \rfloor$
tokenizer
C2S-Scale-Gemma-2-2B
LoRA, rank 16
last-token state h
decision head: $z = Wh + \log b$
prior b: none, training mean, or control cell
softmax: $q \in \Delta^{G-1}$
real cell x (train donor / perturbed)
T
transformation
freq, smooth, log1p, sqrt
target $p = T(x) \in \Delta^{G-1}$
soft CE $-\sum_g p_g \log q_g$
$s \sim$ train library sizes
Mult(s, q) or DM$(s, \kappa q)$
generated cell $\tilde{x}$
score
NLL, cell NLL
MMD, AUC
r_μ, type acc.
UMAP
held-out real cells
inference
conditioning
deterministic measurement
supervision / evaluation